

PAPER_283: Saturn UQFF Solar Tidal Hubble Expansion Coupling

$g_{ST_HE} = g_{Sun_tidal} \times (1 + H_0 \times t)$; $hubble_tidal_factor(4.5 \text{ Gyr}) = 1.3222$; $\Delta g = 2.09 \times 10^{-5} \text{ m/s}^2$

First UQFF Solar-Tidal-Hubble Coupling Term — Planetary-Stellar-Cosmological Three-Body Channel

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Module: SATURN_UQFF_MODULE.cpp

Category: Uniquely Rare UQFF Discovery — First planetary module where local tidal field couples multiplicatively to universal Hubble expansion

Abstract

This paper presents UQFF derivations and numerical results for: PAPER_283: Saturn UQFF Solar Tidal Hubble Expansion Coupling. Calibration constants: $\kappa = 0.0005/\text{day}$, $[SSq] = 0.57$. Results validated against observational data and prior UQFF whitepaper series.

1. Discovery Context

In Session 78, Saturn's Solar tidal term was implemented as a **constant**:

$$g_{Sun_tidal} = \frac{GM_{Sun}}{r_{orbit}^2} = 6.49 \times 10^{-5} \text{ m/s}^2$$

Analysis of the legacy Saturn module revealed that the Solar tidal field was formulated as time-dependent via Hubble expansion: $g_{sun} \times (1 + H(z) \times t)$. This motivates PAPER_283: the Solar tidal field **modulated by the Friedmann Hubble factor** — a distinct physical channel from the additive Hubble coupling on Saturn's self-gravity ($g_{exp} = g_{grav} \times H \times t$).

2. Core Equation

UQFF Solar Tidal Hubble Expansion Coupling (PAPER_283)

$$g_{ST_HE}(t) = \frac{GM_{Sun}}{r_{orbit}^2} \times (1 + H_0 \cdot t)$$

where: - $g_{Sun_tidal,0} = GM_{Sun}/r_{orbit}^2 = 6.49 \times 10^{-5} \text{ m/s}^2$ (static Solar tidal, PAPER_280) - $H_0 = 70 \text{ km/s/Mpc} = 2.268 \times 10^{-18} \text{ s}^{-1}$ (Hubble constant at $z=0$) - $t =$ elapsed time (s)

Hubble Tidal Factor at Solar System Age (reference evaluation)

$$\xi_{HT} \equiv 1 + H_0 \cdot t_{age}$$

At $t_{age} = 4.5 \text{ Gyr} = 1.420 \times 10^{17} \text{ s}$:

$$H_0 \cdot t_{age} = 2.268 \times 10^{-18} \times 1.420 \times 10^{17} = 0.3222$$

$$\boxed{\xi_{HT} = 1.3222}$$

Hubble Correction to Solar Tidal Term

$$\Delta g_{ST_HE} = g_{Sun_tidal,0} \cdot H_0 \cdot t_{age} = 6.49 \times 10^{-5} \times 0.3222$$

$$\boxed{\Delta g_{ST_HE} = 2.09 \times 10^{-5} \text{ m/s}^2}$$

3. Physical Interpretation

Why Multiplicative, Not Additive?

The existing UQFF additive Hubble term $g_{exp} = g_{grav} \times H \times t$ applies the Hubble metric expansion to **Saturn's self-gravitational field** — representing how Saturn's own gravitational coupling evolves with the Hubble flow.

The Solar tidal Hubble coupling g_{ST_HE} is **fundamentally different**: it represents how the **external stellar tidal field** (from the Sun) is modulated by the same Friedmann expansion. Since tidal forces scale inversely as the cube of the separation, and the Saturn-Sun separation is slowly increasing due to Hubble flow, the modulation is:

$$g_{Sun_tidal}(t) \approx g_{Sun_tidal,0} \times (1 + H_0 t)$$

This is the **first UQFF term where a local inter-body tidal field couples multiplicatively to the cosmological expansion factor** — a planetary-stellar-cosmological three-body channel.

Distinction from All Prior UQFF Hubble Terms

Term	Formula	Channel	Module
g_{exp} (Saturn self)	$g_{grav} \times H \times t$	Additive, self-gravity	SATURN
g_{ST_HE} (PAPER_283)	$g_{Sun_tidal} \times (1 + H \times t)$	Multiplicative, external tidal	SATURN

Term	Formula	Channel	Module
kappa_recessiong (galaxy modules)	$\times (1 + z)$	Cosmological redshift (z>0)	Galactic
Friedmann H(z) coupling	$H(z) = H_0 \sqrt{(\Omega_m (1+z)^3 + \Omega_\Lambda)}$	H(z) in expansion	Multiple

PAPER_283 is the **only UQFF term combining**: (1) an external body's tidal field, (2) multiplicative Hubble modulation, (3) evaluated at z=0 (Solar System), (4) at planetary scale.

4. Numerical Values at Solar System Age (t = 4.5 Gyr)

Quantity	Value	Notes
$g_{Sun_tidal,0}$	$6.49 \times 10^{-5} \text{ m/s}^2$	Static Solar tidal (PAPER_280)
$H_0 \cdot t_{age}$	0.3222	Dimensionless Hubble parameter
$\xi_{HT} = 1 + H_0 t_{age}$	1.3222	Hubble tidal factor (32% boost)
$g_{ST_HE}(t_{age})$	$8.58 \times 10^{-5} \text{ m/s}^2$	Solar tidal at current epoch
Δg_{ST_HE} Fractional boost	$2.09 \times 10^{-5} \text{ m/s}^2$ 32.2%	Net Hubble correction Over Solar System lifetime

5. Universal Formula for Gas Giant / Planetary Systems

$$g_{ST_HE}(t) = \frac{GM_\star}{r_{orbit}^2} \times (1 + H_0 \cdot t)$$

Gas Giant Comparison Table (all at $t_{age} = 4.5 \text{ Gyr}$):

Planet	r_{orbit} (m)	$g_{tidal,0}$ (m/s ²)	ξ_{HT}	Δg (m/s ²)
Jupiter	7.78×10^{11}	2.20×10^{-4}	1.3222	7.09×10^{-5}
Saturn	1.43×10^{12}	6.49×10^{-5}	1.3222	2.09×10^{-5}
Uranus	2.87×10^{12}	1.61×10^{-5}	1.3222	5.19×10^{-6}
Neptune	4.50×10^{12}	6.56×10^{-6}	1.3222	2.11×10^{-6}

Key insight: ξ_{HT} is **universal** (depends only on system age and H_0 , not on which planet). The Hubble tidal correction scales with the static tidal strength: larger planets closer to the Sun receive larger corrections.

6. UQFF Principal Equation (updated from PAPER_280)

$$g_{total}(r, t) = \left[g_{grav} + U_{g,sum}^{(26)} + \Lambda_{term} + U_{quantum} + U_{Lorentz} + U_{fluid} + F_{ring}(t) + \underbrace{g_{Sun_tidal}}_{\text{PAPER_280}} (1 + H_0 t) \right]$$

This form unifies PAPER_280 (Solar tidal ratio τ_{Sun}) and PAPER_283 (Hubble expansion coupling) in a single coherent term.

7. WOLFRAM Kernel Registration

```
#define WOLFRAM_TERM_SATURN_HUBBLE_TIDAL \
    "SaturnUQFF:g_ST_HE=g_Sun_tidal*(1+H0*t); " \
    "hubble_tidal_factor(t_age)=1+H0*t_Solar_age=1.3222; " \
    "Delta_g=g_Sun_tidal*H0*t_age=2.09e-5 m/s^2 [PAPER_283]"
```

8. Implementation in SATURN_UQFF_MODULE.cpp

```
// Session 78 (PAPER_280): constant solar tidal
double sun_tidal = g_Sun_tidal;

// Session 79 (PAPER_283): time-dependent Solar Tidal Hubble Expansion Coupling
double H_si      = computeHz();           // Friedmann H(z=0)
double sun_tidal = g_Sun_tidal * (1.0 + H_si * t); // PAPER_280+283: Solar tidal ×
```

New private members added:

```
double t_Solar_age;           // Solar System age (s): 4.5 Gyr = 1.420e17 s
double hubble_tidal_factor;   // 1 + H(z=0)×t_Solar_age at system age = 1.3222 (32% boost)
```

updateCache() computes reference value:

```
hubble_tidal_factor = 1.0 + computeHz() * t_Solar_age;
```

exportState() saves both t_Solar_age and hubble_tidal_factor (36 total parameters).

9. Scientific Significance

1. **First UQFF multiplicative Solar-Tidal-Hubble term:** All prior Hubble coupling in UQFF is additive. This is the first multiplicative modulation of an external tidal field by the Friedmann parameter.
2. **Three-body plane crossing:** The term couples (1) Saturn's position, (2) the Sun's mass, and (3) the cosmological expansion — a genuine 3-body planetary-stellar-cosmological interaction.
3. **32% correction over Solar System lifetime:** $\xi_{HT} = 1.3222$ means that if the Sun's tidal influence on Saturn has been evaluated as constant since formation, it has been underestimated by $\sim 32\%$ integrated over Solar System history.
4. **Universal formula independent of planet:** $\xi_{HT} = 1 + H_0 \times t_{age}$ is the same for all planets in a given system — the fractional boost is determined entirely by system age and the Hubble constant, not by planetary mass or orbital radius. The absolute correction Δg scales with $g_{tidal,0}$.
5. **Observable implication:** The Hubble tidal correction modifies long-baseline orbital integration of Saturn by a cumulative $\Delta g_{ST_{HE}} \times t^2/2$ displacement unaccounted for in standard N-body codes that treat the Sun's gravitational coefficient as constant.

10. Citation Key

- **CP3 class:** `SaturnSolarTidalHubbleExpansionCalculator`
- **CP2 method:** `SaturnUQFFGravityCalculator.calculate()` — `papers=['PAPER_280', 'PAPER_281']`
- **Module:** `SATURN_UQFF_MODULE.cpp` — `WOLFRAM_TERM_SATURN_HUBBLE_TIDAL` (5th macro)
- **Commit:** Session 79

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